

To compute the rejection region and p -value, we need to compute the approximate df for t' :

$$c = \frac{\frac{s_1^2/n_1}{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \frac{(16.37)^2/40}{(16.37)^2/40 + (9.88)^2/40} = .73$$

$$\text{df} = \frac{(n_1 - 1)(n_2 - 1)}{(1 - c)^2(n_1 - 1) + c^2(n_2 - 1)} = \frac{(39)(39)}{(1 - .73)^2(39) + (.73)^2(39)}$$

$$= 64.38, \text{ which is rounded to } 64$$

Table 2 in the Appendix does not have df = 64, so we will use df = 60. In fact, the difference is very small when df becomes large: $t_{.05} = 1.671$ and 1.669 for df = 60 and 64, respectively.

R.R.: For $\alpha = .05$ and df = 60, reject H_0 if $t' > 1.671$.

Since $t' = 3.82$ is greater than 1.671, we reject H_0 . We can bound the p -value using Table 2 in the Appendix with df = 60. With $t' = 3.82$, the level of significance is $p\text{-value} < .001$. Thus we can conclude that there is significant ($p\text{-value} < .001$) evidence that μ_1 is greater than μ_2 . Although we have determined that there is a statistically significant difference between the mean plant densities at the control and oil spill sites, the question remains whether these differences have *practical* significance. We can estimate the size of the difference in the means by placing a 95% confidence interval on $\mu_1 - \mu_2$.

The appropriate 95% confidence interval for $\mu_1 - \mu_2$ is computed by using the following formula with df = 64, the same as was used for the R.R.

$$(\bar{y}_1 - \bar{y}_2 \pm t_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \quad \text{or} \quad (38.48 - 26.93) \pm 2.0 \sqrt{\frac{(16.37)^2}{40} + \frac{(9.88)^2}{40}}$$

$$\text{or} \quad 11.55 \pm 6.05$$

Thus, we are 95% confident that the mean plant densities differ by an amount between 5.5 and 17.6. The plant scientists would then evaluate whether a difference in this range is of practical importance.

Reporting Conclusions We would need to write a report summarizing our findings from the study. The report would include the following items:

1. Statement of the objective for the study
2. Description of the study design and data collection procedures
3. Numerical and graphical summaries of the data sets
 - Table of means, medians, standard deviations, quartiles, range
 - Boxplots
 - Stem-and-leaf plots
4. Description of all inference methodologies:
 - Approximate t tests of differences in means
 - Approximate t -based confidence interval on population means
 - Verification that all necessary conditions for using inference techniques were satisfied using boxplots, normal probability plots